Moynuddin Al Masum

AI Engineer (Computer Vision)

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Social Info

LinkedIn
GitHub
LeetCode
Portfolio

Skills: Generative AI, Computer Vision, ML, RAG, LLMs, LangChain, n8n, FastAPI, NLP, TensorFlow, PyTorch, Scikit-learn, Neural Networks, Python, C++, Edge AI (DEEPX NPU), Docker, React

About Me

I work with Machine Learning, Deep Learning, and AI, using tools like TensorFlow, PyTorch, Scikit-learn, Python, C++, OpenCV, YOLOv8, ByteTrack, Keras, Pandas, and Matplotlib to build real-time, intelligent solutions. My projects include AI-based surveillance, fire & smoke detection, and automated attendance systems, integrating computer vision and NLP. As an AI Engineer (Computer Vision) at SmartData Technologies, I deploy real-time vision systems on the DEEPX DX-M1 NPU. I have hands-on experience in end-to-end model development, optimization, and deployment.

Education

Bangladesh University of Professionals
BSc in Information & Communication Engineering, Jun 2021 - Jun 2026

Experience

AI Engineer (Computer Vision) 1 Sep 2026 – Present
SmartData Technologies Limited

- Build and deploy real-time computer-vision systems on the DEEPX DX-M1 NPU — face-recognition attendance, hall management, conveyor bag counting and multi-tenant office analytics — using C++, GStreamer (VA-API GPU decoding) and Docker.
- Re-engineered the BUP face-attendance pipeline into a C++ NPU engine (rank-1 accuracy 92.8% → 99.0%, 1.7× throughput, 3.1× less CPU) and built the BUP Smart Hall, CCPLC bag-counting and Smart Office analytics platforms end to end.

AI Engineer Intern 1 May 2026 – 30 Aug 2026 (4 months)
SmartData Technologies Limited | Internship Certificate

- Completed a 4-month internship in Artificial Intelligence Engineering (computer vision), then continued as a full-time AI Engineer from 1 September 2026.

Projects

Real-Time AI Home Security System: *Link*
Built a real-time home surveillance system that detects unknown persons via webcam/IP camera, identifies family members using InsightFace face recognition, triggers an audible alarm, and delivers instant Telegram notifications with intruder photos for rapid security response.

- Tools Used: InsightFace, OpenCV, Telegram Bot API, Real-time Video Processing, Python, Multithreading.

Automatic Attendance System for University: *Video*
Built a real-time classroom attendance system for Bangladesh University of Professionals: IP-CCTV streams are processed by a C++ engine (SCRFD face detection + ArcFace recognition) on the DEEPX DX-M1 NPU, marking students present automatically and generating attendance reports and absent lists. Re-engineering the original Python pipeline raised rank-1 identification from 92.8% to 99.0% (318 enrolled photos) with 1.7× higher throughput at 3.1× lower CPU; shipped as a Docker image with role-based web dashboards.

- Tools Used: C++, pybind11, DEEPX DX-M1 NPU (DXRT), SCRFD, ArcFace, OpenCV, GStreamer, RTSP / IP-CCTV, Flask, Docker.

Smart Office – AI Office Analytics Platform: *Video*
Built a multi-tenant office-analytics platform that runs three models (YOLO26-pose, SCRFD-10G, ArcFace) in one C++ binary on the DEEPX DX-M1 NPU: face-recognition attendance, restricted-zone, fall-detection, unauthorised-person and desk on-duty modules, with Telegram and e-mail alerts that carry the photo. One engine per NPU core raised throughput 2.6× (71.8 → 186.4 inferences/s) and software decoding measured 33× faster than the board's hardware decoder; tenant-isolated JWT API, React dashboard and PostgreSQL in Docker with 143 automated tests.

- Tools Used: C++17, DEEPX DX-M1 NPU (DXRT), YOLO26-pose, SCRFD, ArcFace, GStreamer, Drogon, PostgreSQL, React, Docker.

Real-time Object Detection & Smart Counting System: *Link* | *Video*
Built a real-time IP CCTV camera–based video analytics system for live object detection, multi-object tracking, and bi-directional IN/OUT counting from RTSP streams. Detects students and multiple vehicle types with stable ID tracking and optimized GPU performance for production-ready monitoring.

- Tools Used: Ultralytics YOLOv8, Custom YOLO Model, OpenCV, CCTV RTSP Stream Integration, ByteTrack, TensorRT

Smart Hall Management System: *Video*
Built a management platform for the residential halls of Bangladesh University of Professionals as a single C++20 server in Docker. Face-recognition gate (entry/exit) and dining cameras run on the DEEPX DX-M1 NPU alongside meal planning, a meal counter, kitchen board, complaints, notice board and automatic monthly meal bills. SCRFD + ArcFace (INT8, compiled with DX-COM) with ByteTrack identity tracking reach 99.9% first-match accuracy on 133 enrolled students (1,907 photos), 0% non-enrolled people accepted, 11 ms per face.

- Tools Used: C++20, DEEPX DX-M1 NPU, SCRFD-10G, ArcFace, ByteTrack, GStreamer (VA-API), SQLite, RTSP, Docker.

CCPLC – AIoT Cement Bag Counting System: *Video*
Built a real-time bag-counting and loading-management system for conveyor belts: IP-camera streams are decoded on the edge (GPU / VA-API) and run through a YOLOv8s model on the DEEPX DX-M1 NPU (~197 fps capacity); bags are tracked and counted when they enter an operator-drawn 3-D zone, then booked into loading sessions with PCC/OPC targets, shift reports (CSV/PDF) and LED count displays. Includes a React control-room dashboard and a C++17 backend, deployed with Docker.

- Tools Used: YOLOv8, DEEPX DX-M1 NPU, C++17, OpenCV, GStreamer (VA-API), React, Vite, WebSocket / REST API, nginx, Docker.

Blind Navigation Hybrid AI System: *Link*
Developed a real-time blind-assistance system using Mobile-App based object detection that identifies roads, obstacles, potholes, stairs, and walkable paths while providing continuous audio navigation guidance toward the destination.

- Tools Used: YOLOv8, Real Time Custom Object Detection, OpenCV, MiDaS/ZoeDepth, Python, STT, TTS, Speech Recognition, Real-time Video Processing.

BUP RAG Chatbot: *Link* | *Video*
Developed an intelligent FAQ chatbot for Bangladesh University of Professionals (BUP) using a Retrieval-Augmented Generation (RAG) pipeline that semantically searches FAQs with FAISS and generates accurate, language-adaptive answers via Google Gemini API in real time.

- Tools Used: FAISS, SentenceTransformer, Google Gemini API, Gradio, RAG (Retrieval-Augmented Generation)

AI LinkedIn Post Generator: *Link* | *Video*
Developed an AI-powered tool that generates professional, structured LinkedIn posts in multiple languages based on a given topic, using a user-friendly web interface.

- Tools Used: LangChain, OpenAI GPT-4o-mini, Gradio, NLP, Prompt Engineering

Publications (Under Review)

- Multi-Agent Framework for Pediatric Pneumonia Diagnosis Integrating Vision and Language Models (**Accepted**)
- A Hybrid Deep-Learning Framework with Interpretable Risk Scoring for Early Gestational Diabetes Prediction